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Full length article



Modes of failure of viscoplastic rectangles, triangles and drums☆

J. Dai [a,*], N.J. Balmforth [a], D.R. Hewitt [b]

[a] Department of Mathematics, University of British Columbia, Vancouver, BC, V6T 1Z2, Canada
[b] Department of Applied Mathematics and Theoretical Physics, University of Cambridge, Wilberforce Road, Cambridge CB3 0WA, UK

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ABSTRACT

A block of yield-stress fluid fails under gravity when its surface becomes too steep. In this study, we examine the modes of failure of rectangular and triangular blocks leaning against a vertical wall, and of a partially filled cylindrical drum. We present computational results for the failure modes using an augmented-Lagrangian scheme. For rectangular blocks leaning against a wall, the computations demonstrate multiple possible modes of failure depending on the block's geometry. The numerical results for two particular modes of failure match corresponding slipline solutions from the slipline analysis of plasticity theory. The failure modes of triangular blocks are more complicated, although one mode of failure can be straightforwardly described by slipline theory. For both rectangles and triangles, the failure criterion can be constructed asymptotically in the slender limit, and approximately using simple lower-bound calculations for general aspect ratios. For the drum, the computational results show that failure arises along a circular arc, with overlying material rotating rigidly out of position. Again this mode of failure can be described analytically.

1. Introduction

One of the key properties of a viscoplastic fluid is its yield stress: the material deforms like a solid when stresses remain relatively small, but then flows like a fluid if a certain threshold in stress is breached. A number of problems involving viscoplastic materials focus on situations in which the forces imposed on the material can be controlled and slowly raised, to trigger failure events in which the solid-like state collapses when the yield stress is first surpassed; we refer to the ensuing flow pattern as the "failure mode". Such modes characterize incipient failure, and must stitch together a patchwork of unyielded solid-like plugs with flowing regions where the yield point has been reached first. At the moment of failure, the associated velocity field necessarily remains small, and so the failure mode is controlled primarily by the rate-independent plastic stresses. That is, the failure mode is a characteristic of the perfectly plastic limit of the problem.

From a mathematical perspective, the onset of motion in a simpler medium is normally approached using linear stability theory; the most dangerous normal modes are those that grow once some stability threshold is exceeded. For a yield-stress fluid, the situation is different because the nonlinearity of the constitutive law renders any linear stability analysis irrelevant when the basic state is not moving (an infinitesimal perturbation being unable to break the yield stress). Instead, our failure modes correspond to the solutions of a nonlinear eigenvalue problem: the eigenvalue is a dimensionless parameter gauging the strength of gravity to the yield stress (the Oldroyd number Od), and takes a specific value at the onset of motion; the failure mode is the corresponding eigenfunction that satisfies the nonlinear flow problem.

In the current paper, we study the failure of a viscoplastic fluid under its own weight, which is a common component of a variety of applied problems, ranging from geophysical landslides to food processing and simple rheometric tests [1–4]. With a prescribed shape for the viscoplastic fluid, failure usually occurs when a geometrical length reaches a critical value, corresponding to the instance that the imposed gravitational stresses reach the yield point. Above this threshold, the fluid collapses and runs out to a final different shape. Our interest here lies in the initial failure mode, not the final shape at which flow is arrested. Our study is therefore aligned with the classical limit analyses of plasticity theory and geotechnical engineering [5–7].

More specifically, here we consider failure modes of two-dimensional incompressible, viscoplastic fluid, ignoring any solid-like deformation below the yield stress. We explore three particular configurations, as shown in Fig. 1: in the first, a rectangle is set against a vertical wall on top of a horizontal plane. This configuration is the setting for the classical dambreak problem, in which a reservoir of fluid is suddenly released by the removal of an obstructing dam at one end. When the fluid has a yield stress, the material will only collapse if the

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* Corresponding author.
E-mail address: jiyuandai@math.ubc.ca (J. Dai).

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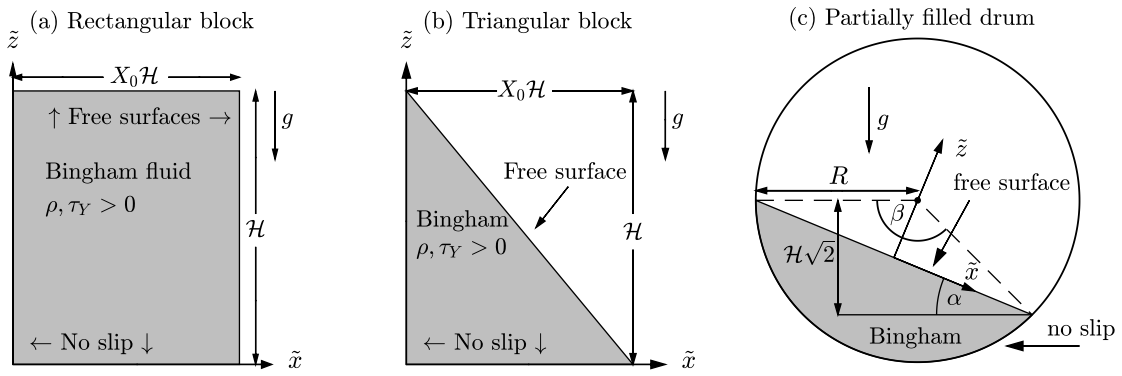


Fig. 1. Dimensional sketch of the geometry, for (a) a rectangular block, (b) a triangle, and (c) a partially filled drum with free surface inclination $\alpha = \frac{3\pi}{8}$ and angular coverage $\beta = \frac{3\pi}{4}$ (fill fraction of $f \approx 26.2\%$).

reservoir is sufficiently deep. The critical depth must depend on the width of the reservoir, except when the rectangle becomes sufficiently wide that fluid only collapses in the vicinity of the original dam. This problem has received attention in the geotechnical engineering literature, in connection with the failure of an embankment [8,9], and in rheology, where it offers a means for practical rheometry [3,4]. Previous discussions have either considered either infinitely wide configurations (embankments, in which collapse is localized to the cliff; [8,9]), or considered the symmetrical failure of a block [10,11]. Here, we explore how the failure mode is impacted when the fluid is set against a no-slip wall, as in any experimental realization of the problem.

Our second configuration sets a triangle against the wall, and is once more emplaced above a horizontal surface. This configuration again has practical application to when fluid is placed inside a relatively wide tank that is tilted about one of its corners. Flattening the fluid surface, and then tilting the tank back to its originally horizontal position, furnishes a triangle that collapses if sufficiently tall. In this case, the critical height is again expected to depend on aspect ratio. Experiments using various viscoplastic fluids in this configuration were reported by [12].

Finally, we also consider the failure of a viscoplastic fluid inside a partially filled drum. This configuration also has an experimental counterpart, in which fluid can be placed in a relatively long cylinder, up to a certain level, and then slowly rotated up to failure. Partially filled drums are popular devices to study the flow of granular media [13]. In particular, for low rotation rates, the granular medium undergoes a succession of episodic avalanches in which sudden, relatively swift collapses punctuate phases of solid-like rigid rotation (*e.g.* [14–16]).

In Section 2, we formulate the three failure problems for an incompressible, two-dimensional fluid with negligible inertia. We employ the Bingham constitutive law, even though its constant plastic viscosity is not reflective of any real fluid, primarily for simplicity and because our main focus is on the plastic limit (in which the rate-dependent part of the stress plays a less significant role). In this section, we also highlight some useful tools from plasticity theory [6,7] that assist our analysis. Following on, we describe our results in Section 3, and then conclude in Section 4. The Appendices contain a number of technical details underscoring our analysis.

2. Formulation

2.1. Governing equations

Consider a two-dimensional steady, incompressible, inertialess Bingham fluid subject to gravity. As sketched in Fig. 1, for each of our three configurations, we describe the geometry using the Cartesian coordinates $\tilde{\mathbf{x}} = (\tilde{x}, \tilde{z})$, with the fluid velocity $\tilde{\mathbf{u}} = (\tilde{u}, \tilde{w})$. Conservation of mass and momentum are given by

$$\nabla \cdot \tilde{\mathbf{u}} = 0, \tag{2.1}$$

$$0 = \nabla \cdot \tilde{\boldsymbol{\tau}} - \nabla \tilde{p} - \rho \mathbf{g}, \tag{2.2}$$

where $\tilde{p}$ is the pressure, $\tilde{\boldsymbol{\tau}}$ is the deviatoric stress, ρ is the fluid density, and $\mathbf{g}$ is gravitational acceleration. The Bingham constitutive law is given by

$$\begin{cases} \tilde{\dot{\gamma}}_{ij} = 0, & \tilde{\tau} < \tau_Y, \\ \tilde{\tau}_{ij} = \left(\mu + \frac{\tau_Y}{\tilde{\dot{\gamma}}}\right)\tilde{\dot{\gamma}}_{ij}, & \tilde{\tau} \geq \tau_Y, \end{cases} \tag{2.3}$$

where τ_Y is the yield stress, μ is the plastic viscosity,

$$\tilde{\dot{\gamma}}_{ij} = \frac{\partial \tilde{u}_i}{\partial \tilde{x}_j} + \frac{\partial \tilde{u}_j}{\partial \tilde{x}_i}, \tag{2.4}$$

and the second tensor invariants are defined as

$$\tilde{\tau} = \sqrt{\frac{1}{2}\sum_{i,j} \tilde{\tau}_{ij}\tilde{\tau}_{ij}}, \quad \tilde{\dot{\gamma}} = \sqrt{\frac{1}{2}\sum_{i,j} \tilde{\dot{\gamma}}_{ij}\tilde{\dot{\gamma}}_{ij}}, \tag{2.5}$$

where $(\tilde{x}_1, \tilde{x}_2) \equiv (\tilde{x}, \tilde{z})$ and $(\tilde{u}_1, \tilde{u}_2) \equiv (\tilde{u}, \tilde{w})$. For all three shapes, we impose no-slip along the contact with the solid surfaces and neglect any interfacial forces along the free surfaces.

2.2. Scaling

To nondimensionalize the equations, we scale lengths by a representative change in elevation along the free surface, $\mathcal{H}$. We then scale the pressure and stress by a characteristic hydrostatic pressure $\rho g \mathcal{H}$, and define a dimensionless yield stress by

$$\mathrm{Od} = \frac{\tau_Y}{\rho g \mathcal{H}}. \tag{2.6}$$

The precise definition of $\mathcal{H}$ and the remaining scaling of the problem depend on the choice of configurations, as outlined below.

2.2.1. Rectangles and triangles

For the rectangular and triangular geometries, we select $\mathcal{H}$ to be the height of the block, as indicated in Fig. 1(a,b). The aspect ratio X_0 that compares the width at the bottom to the height becomes a parameter of both problems. To scale the velocity components, we use the characteristic speed,

$$\mathcal{U} = \frac{\rho g \mathcal{H}^2}{\mu}. \tag{2.7}$$

Altogether, we now define the dimensionless variables,

$$(x, z) = \frac{(\tilde{x}, \tilde{z})}{\mathcal{H}}, \quad \dot{\boldsymbol{\gamma}} = \frac{\mathcal{H}}{\mathcal{U}}\tilde{\dot{\boldsymbol{\gamma}}}, \tag{2.8}$$

$$(u, w) = \frac{(\tilde{u}, \tilde{w})}{\mathcal{U}}, \quad (p, \boldsymbol{\tau}) = \frac{(\tilde{p}, \tilde{\boldsymbol{\tau}})}{\rho g \mathcal{H}}. \tag{2.9}$$

The dimensionless versions of the model equations are given below in Section 2.2.3; the main changes arise in the constitutive law, wherein the dimensional viscosity $\mu + \tau_Y/\tilde{\dot{\gamma}}$ becomes replaced by $1 + \mathrm{Od}/\dot{\gamma}$, and in the force of gravity, which becomes replaced by the unit vector, $\hat{\mathbf{e}}_g = -\hat{\mathbf{z}}$.

2.2.2. Drums

For the drum, we prepare the model equations slightly differently: the goal is to scale and re-orientate so that free surface has a length of 2 and is inclined with respect to gravity by 45°. The first step is to align the Cartesian coordinate system with the original free surface, which has slope $\tan\alpha$ (see Fig. 1). For definiteness, we consider free surfaces that slope downwards to the right (in the positive $x-$direction). We then scale lengths by $R\sin\frac{1}{2}\beta$, where R is the original drum radius, and velocity by the speed scale

$$\mathcal{U} = \frac{\rho g R \mathcal{H}}{\mu}. \tag{2.10}$$

The length $\mathcal{H}$ is now chosen to be the net elevation change along the free surface, scaled by $1/\sqrt{2}$; see Fig. 1. That is,

$$\mathcal{H} = \sqrt{2} R \sin\alpha \sin\tfrac{1}{2}\beta. \tag{2.11}$$

In terms of drum radius, our Oldroyd number is then,

$$\mathrm{Od} = \tau_Y \left[\sqrt{2}\rho g R \sin\alpha \sin\tfrac{1}{2}\beta\right]^{-1}. \tag{2.12}$$

At this stage, the gravitational force, scaled by $\rho g\mathcal{H}/(R\sin\frac{1}{2}\beta)$, becomes

$$\frac{R\sin\frac{1}{2}\beta}{\mathcal{H}} \begin{pmatrix} \sin\alpha \\ -\cos\alpha \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -\cot\alpha \end{pmatrix} \tag{2.13}$$

Last, we define the dimensionless pressure p to be

$$p = \frac{\tilde{p}}{\rho g \mathcal{H}} + \frac{1}{\sqrt{2}} z(\cot\alpha - 1). \tag{2.14}$$

Because the free surface is located at $z = 0$, the additional hydrostatic component in p does not affect the stress-free boundary conditions, and merely re-orientates the dimensionless gravity force to

$$\hat{\mathbf{e}}_g = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}. \tag{2.15}$$

Finally, bearing in mind the different choice for scaling lengths (*i.e.* $R\sin\frac{1}{2}\beta$ rather than $\mathcal{H}$), we replace (2.8) with

$$(x, z) = \frac{(\tilde{x}, \tilde{z})}{R\sin\frac{1}{2}\beta}, \quad \dot{\boldsymbol{\gamma}} = \frac{R\sin\frac{1}{2}\beta}{\mathcal{U}} \tilde{\dot{\boldsymbol{\gamma}}}, \tag{2.16}$$

but retain (2.9). Again, we postpone quoting the dimensionless equations to Section 2.2.3. In addition to Od, which encodes the original tilt angle α, the drum has the fill fraction f as a model parameter (or, equivalently, the angle β in Fig. 1, with $f = (\beta - \sin\beta)/(2\pi)$).

2.2.3. Dimensionless model

For all three configurations, our dimensionless model equations now take the form,

$$\nabla \cdot \mathbf{u} = 0, \tag{2.17}$$

$$0 = \nabla \cdot \boldsymbol{\tau} - \nabla p - \hat{\mathbf{e}}_g, \tag{2.18}$$

$$\begin{cases} \dot{\gamma}_{ij} = 0, & \tau < \mathrm{Od}, \\ \tau_{ij} = \left(1 + \frac{\mathrm{Od}}{\dot{\gamma}}\right)\dot{\gamma}_{ij}, & \tau \geq \mathrm{Od}. \end{cases} \tag{2.19}$$

We numerically solve these equations (along with the respective boundary conditions) using an augmented Lagrangian scheme for general values of Od and X_0 or f. Note that, because we have discarded inertia, the computed solution provides a non-zero snapshot of the velocity field whenever the fluid yields somewhere; the fluid does not accelerate from rest. However, our main focus is on the failure modes for which the velocity field becomes arbitrarily small, corresponding to the onset of motion. To find these modes, a search is made varying Od, monitoring the magnitude of the velocity field and heading in the direction in which this decreases. An extrapolation to $\mathrm{Max}(|\boldsymbol{u}|) \to 0$ then furnishes the critical value Od_c, and the closest solution is used to identify the stress and velocity fields of the corresponding failure model. The details of the full numerical implementation are described in Appendix A.

2.3. Plasticity theory

In the limit in which motion first begins, and the dimensionless yield stress Od approaches a certain threshold Od_c, the velocity becomes small and the plastic stress dominates over the viscous stress except over any narrow shear layers. For this "perfectly plastic limit", the constitutive relation reduces to

$$\tau_{xx}^2 + \tau_{xz}^2 = \mathrm{Od}^2. \tag{2.20}$$

Following standard developments in plasticity theory [6,7], we then set

$$(\tau_{xx}, \tau_{xz}) = \mathrm{Od}(-\sin 2\theta, \cos 2\theta) \tag{2.21}$$

and

$$P = p - \hat{\mathbf{e}}_g \cdot \begin{pmatrix} x \\ z \end{pmatrix}. \tag{2.22}$$

The force balance equation (2.18) then simplifies to a two-dimensional hyperbolic system:

$$\begin{aligned} &\left(\tan\theta\frac{\partial}{\partial x} + \frac{\partial}{\partial z}\right)(P + 2\mathrm{Od}\,\theta) = 0, \\ &\left(-\cot\theta\frac{\partial}{\partial x} + \frac{\partial}{\partial z}\right)(P - 2\mathrm{Od}\,\theta) = 0. \end{aligned} \tag{2.23}$$

There are therefore two families of orthogonal characteristics, the sliplines of plasticity theory. Details of some explicit constructions of the slipline field for rectangular and triangular blocks are provided in Appendix C.

2.4. Power balance and bounds

In the perfectly plastic limit, the power balance,

$$\mathrm{Od} = \frac{\mathcal{P}}{\mathcal{E}}, \tag{2.24}$$

can be derived from the governing equations. Here,

$$\mathcal{P} = -\iint_D w \,\mathrm{d}x\mathrm{d}z \tag{2.25}$$

denotes the potential energy released by falling,

$$\mathcal{E} = \iint_D \dot{\gamma} \,\mathrm{d}x\mathrm{d}z \tag{2.26}$$

is the scaled plastic dissipation rate, and D denotes the fluid domain.

In the limit analysis of plasticity theory, this power balance is embedded in a variational formulation from which one can extract lower bounds on the critical threshold Od_c. For this task, one constructs trial incompressible velocity fields that do satisfy the velocity boundary conditions, but do not satisfy either the constitutive law or the force balance equations. The lower bound is given by maximizing $\mathcal{P}/\mathcal{E}$ over all parameters in the trial velocity field. Some examples are given in Appendices E and F.

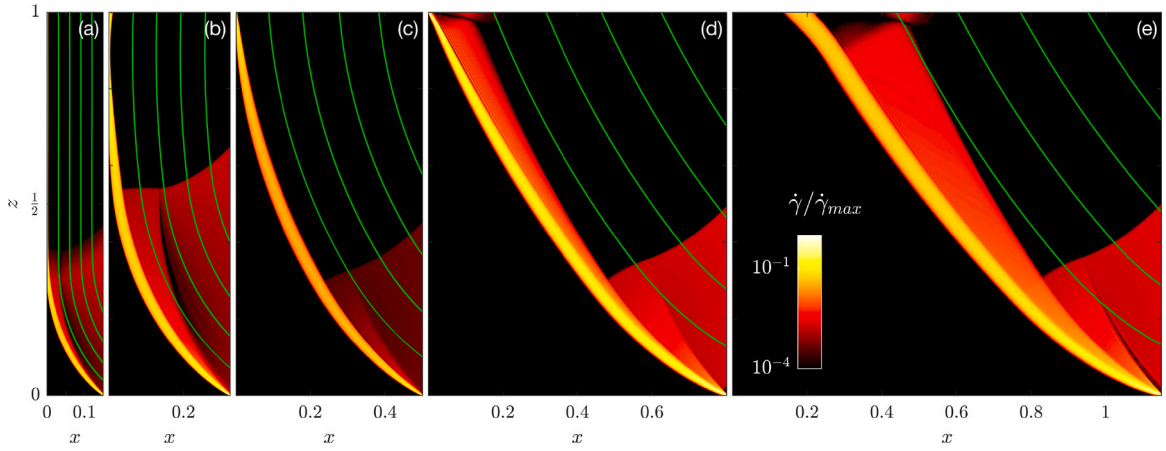


Fig. 2. Failure modes for rectangles with (a) $X_0 = 0.15$, (b) $X_0 = 0.325$, (c) $X_0 = 0.5$, (d) $X_0 = 0.8$, (e) $X_0 = 1.15$. Shown are selected streamlines superposed on logarithmic density maps of strain rate $\dot{\gamma}$.

3. Results

3.1. Rectangles

Sample solutions close to the plastic limit for rectangles with varying aspect ratio X_0 are displayed in Fig. 2. Shown are density plots of the shear-rate invariant $\dot{\gamma}$ along with a selection of superposed streamlines. The shear-rate maps highlight three distinctive subregions within the overall flow patterns: the black regions indicate plugs. The slender bright yellow contours trace out localized layers of strong shear, which have finite thickness for finite Bingham number but thin into discontinuous lines of failure in the plastic limit [17,18]. The more extended, redder regions identify the areas that yield to support almost perfectly plastic deformation.

The figure demonstrates that, as X_0 increases, the rectangles switch from tall columns that fail near their base to wider blocks that only collapse around the top right-hand corner. Altogether, five different types of flow pattern can be identified as X_0 increases, with the five panels of Fig. 2 illustrating these different failure modes; we refer to these cases as modes $(i) - (v)$. Although the flow patterns and their progression with aspect ratio have similarities to those observed when free-slip or symmetry conditions are applied along the left-hand wall ($x = 0$) [11,18], the details are somewhat different.

More specifically, for narrow columns and mode (i), the no-slip condition at $x = 0$ demands that failure must also arise all the way up that wall, unlike in the free-slip case. For the solution shown in Fig. 2(a), a large fraction of the upper part of the column falls rigidly and vertically, and a thin boundary layer with high shear rate forms along the wall. Underneath the descending plug, a zone of plastic deformation appears, buffered by a free shear layer from another, stagnant plug that locks the lower left corner of the block into place. The shear layer is the continuation of the wall boundary layer, which breaks free of the side wall and curves to the right to eventually meet the lower right corner of the block. Without excessive magnification of panel (a) in Fig. 2, it is hard to see the boundary layer along the left wall (the boundary layer is more visible in the example given in Fig. 3).

The geometry of mode (i) can be captured by slipline theory, in a modification of that used to describe failure with a free-slip wall or symmetry line along $x = 0$ [10,11]. The construction consists of a curved checkerboard of sliplines that leave the free surface along the right-hand border of the block, supplemented by a fan emerging from the lower right corner (*cf.* Appendix C). The slipline pattern for $X_0 = \frac{1}{5}$ is shown in Fig. 3, and compared to a corresponding numerical solution. The main difference with the free-slip case is that the slipline field is terminated to the left by a slipline from the fan that becomes vertical on reaching the left wall. The second slipline that leaves this

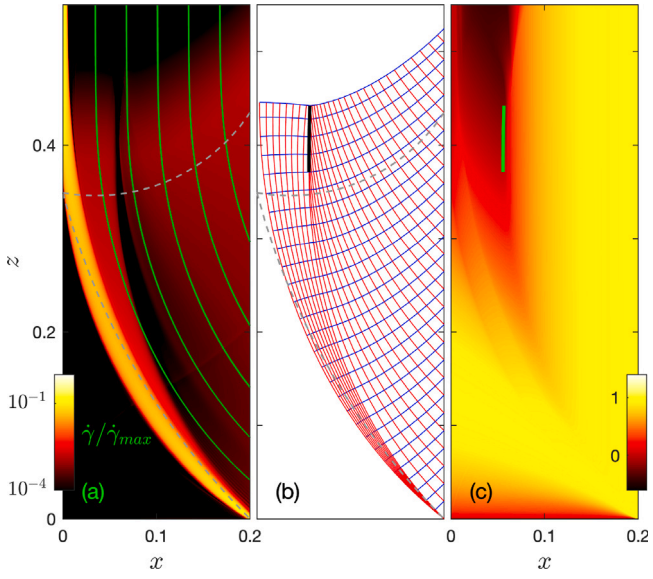


Fig. 3. Further details of failure mode (i) for a rectangle with $X_0 = \frac{1}{5}$, showing (a) selected streamlines superposed on a logarithmic strain rate density plot, (b) the corresponding slipline construction, and (c) τ_{xx}/Od. The dashed lines in (a,b) show the results of the lower bound calculation (with a combined vertical-rotational configuration); the thicker solid line in (b,c) indicates the stress discontinuity predicted by slipline theory.

position furnishes the top border of the plastic zone. An unavoidable consequence of the left-hand boundary condition (*i.e.* that the leftmost slipline becomes vertical) is that sliplines must begin to cross and form a stress discontinuity along the border between the fan and checkerboard below the uppermost slipline. The location of this discontinuity is highlighted in panel (b) of Fig. 3, and panel (c) illustrates how the feature becomes embedded in the stress component τ_{xx} of the numerical solution. Because the discontinuity is almost vertical, slipline theory demands that τ_{xz} remains continuous across this line, but τ_{xx} develops a jump on ascending that feature (as shown).

For wider columns, the additional resistance provided by the left wall eventually implies that it becomes energetically preferable for falling fluid to completely break free of that wall for some critical aspect ratio, and then rotate to the right during the fall. This leads to solutions like those shown in Fig. 2(b,c), where the upper plugs now rotate rather than fall purely vertically. The difference between these two solutions lies in the centre of rotation: the centre is below the top

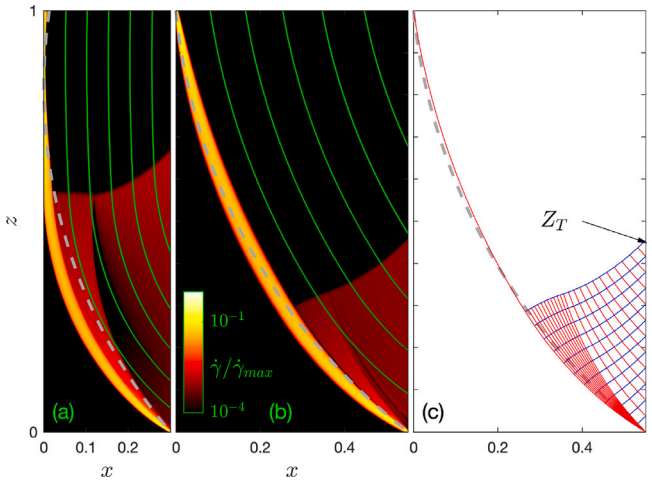


Fig. 4. Further details of failure modes (ii) and (iii) for rectangles with (a) $X_0 = 0.35$ and (b,c) 0.55, respectively. Panels (a,b) display selected streamlines superposed on a logarithmic strain rate density plot; the corresponding slipline construction for $X_0 = 0.55$ is shown in (c). The dashed lines show the results of the lower bound calculation using a single rotational configuration.

of the rectangle in Fig. 2(b), and above it in Fig. 2(c). In both cases, the upper plug is bordered to the left by a circular arc of failure. For the first solution, an example of mode (*ii*), the arc grazes the left wall; for the second, mode (*iii*), the arc intersects the top left corner of the block.

Again, these solutions can be characterized by a slipline field, as illustrated in Fig. 4. Somewhere along its length, the leftmost slipline now reaches the top of the slipline field and turns into the circular failure arc. Note that, because this slipline does not then need to become vertical, it is no longer necessary for a stress discontinuity to disfigure the slipline pattern for modes (*ii*) and (*iii*). Nevertheless, for mode (*ii*), it turns out that the circular failure arc always becomes nearly vertical (see Fig. 4(a)), and it is then difficult to distinguish the slipline field of this mode from that of mode (*i*).

Another awkward feature of mode (*ii*) is that the circular arc cannot meet the top surface of the block at 45°, which is the translation of the stress-free condition to the slipline field (as evident for the checkerboard stemming from the free surface to the right). Consequently, a small plastic region must inevitably appear around the intersection of the circular failure arc with $z = 1$. This small region is hard to see in Fig. 2(b), and implies a change in direction of the shear layer as well as an adjustment to the geometry of the flow pattern. To avoid modifying the slipline construction to take into account this awkward feature, we do not construct in detail the slipline field for mode (*ii*).

For higher aspect ratio still, the top plug does not remain fully intact, but a further plastic zone develops on the left, generating mode (*iv*). Eventually, in mode (*v*), the shear layer detaches from the top left corner, leaving an unyielded central core for $X_0 > 1$. For such aspect ratios, the boundary conditions applying on the left wall become irrelevant, the intervening plug cloaking their effects. The solutions for mode (*v*) are then identical to the corresponding solutions imposing free slip at $x = 0$. The slipline field for these solutions was given previously by Martin [19].

Diagnostics summarizing certain details of the five failure modes are shown in Fig. 5. The critical yield stress Od_c is plotted as a function of X_0 in Fig. 5(a). This threshold increases monotonically with X_0, following the line $\mathrm{Od}_c \sim X_0$ for $X_0 \ll 1$, and converging to the constant value $\mathrm{Od}_c \approx 0.264$ for $X_0 > 1$. The first limit follows from an asymptotic analysis for $X_0 \to 0$ (see Appendix D); the second limit is given by Martin's slipline construction.

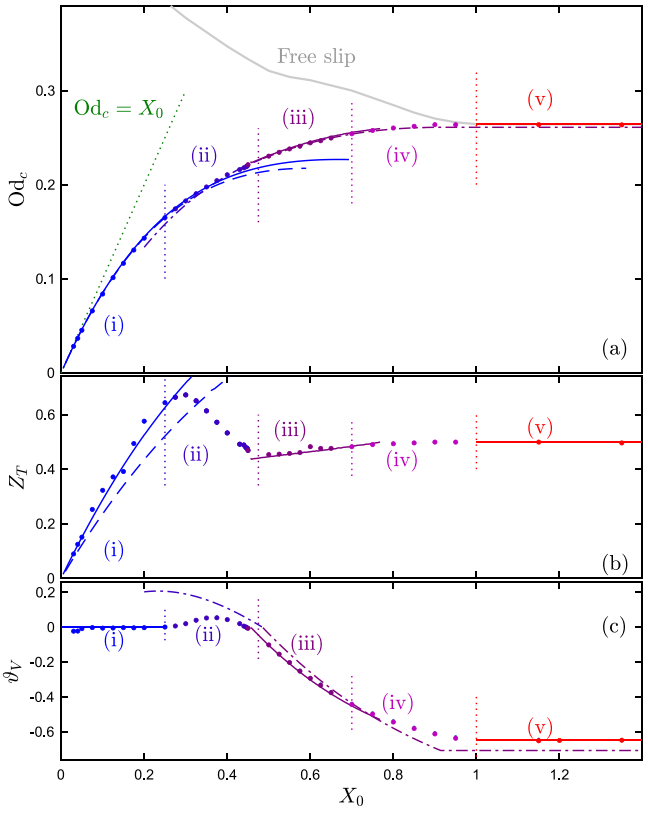


Fig. 5. Diagnostics of failure modes for rectangles: (a) Od_c, (b) the top of the yielded region along the right-hand side of the block, Z_T, and (c) the velocity angle $\vartheta_V = \tan^{-1}(u/w)$ at the top right corner. The dots show numerical results, the solid lines indicate slipline calculations, and the dashed–dotted and dashed lines show lower bounds (with combined vertical-rotational and single rotational configurations, respectively; see Fig. 6). The plots are divided into the five modes identified, and colour-coded accordingly. In (a), the threshold in the slender limit, $\mathrm{Od}_c \sim X_0$, is indicated in green. The grey line shows the results from [11] for rectangles with free slip boundary conditions at $x = 0$.

The two other diagnostics shown in Fig. 5 correspond to the height Z_T attained by the yielded region up the right side and angle that the velocity of the top right corner makes with the vertical, $\vartheta_V = \tan^{-1}(u/w)$. These two diagnostics help characterize the differences between the five failure modes. For example, in mode (*i*), the top of the block falls vertically, and so $\vartheta_V = 0$. By contrast in mode (*ii*), $\vartheta_V > 0$ because the rotation centre is below $z = 1$; for the other modes, $\vartheta_V < 0$.

Also included in Fig. 5 are the predictions of the slipline constructions for modes (*i*) and (*iii*), and the earlier results of [11] for rectangles with free slip at $x = 0$. The latter have thresholds Od_c that always lie above Martin's limit, whereas our no-slip results fall below (both features are due to the differing degrees of resistance from the left wall).

In this figure, we further include results from simple lower-bound calculations. These specific lower bounds follow from considering the trial velocity configurations shown in Fig. 6. The case in panel (a) shows a combined vertical and rotational configuration with arcs of failure along the curves shown by thicker red lines; more details of the construction can be found in [11]. Those in (b,c,d) are similar to constructions used previously in geotechnical engineering [8,9] and contain only a single rotational configuration, constrained in different ways by the walls. Each case contains one or more angles that act as optimization parameters (as indicated). For slender columns, configuration (a) performs better than the others, but not for $X_0 \gtrsim 0.34$. The lower bound results compare quite well with the numerical results for both the threshold Od_c and other diagnostics, although $z_T = 0$

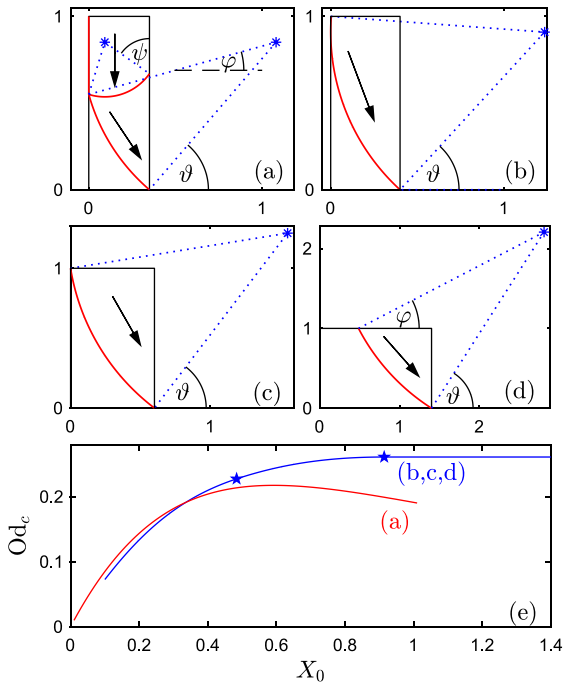


Fig. 6. Lower bound configurations. Sketched are (a) the combined vertical/rotational configuration and (b,c,d) the pure rotational configuration. The thicker (red) lines show the lines and arcs of failure, with the dotted lines indicating the rotation centres. The optimization angles are indicated, and the stars indicate the centres of rotation and the upper failure arc in (a). In (e), the results are collected together into a plot of Od_c against X_0. The stars indicate the switches between the configurations in (b–d).

for the purely rotational configuration. The failure arcs for configuration (a) and $X_0 = \frac{1}{5}$ are included in Fig. 3; those for configuration (b,c) with $X_0 = 0.35$ and 0.55 in Fig. 4. Qualitatively, at least, the lower bounds are relatively successful in reproducing features of the numerical solutions.

3.2. Triangles

A gallery of failure modes computed for triangles with different aspect ratio X_0 are displayed in Fig. 7. As shown in Appendix B, a transformation of coordinates can be introduced to demonstrate that solutions for triangles with aspect ratio X_0 and X_0^{-1} are equivalent. We therefore consider only the range $0 < X_0 \leq 1$. Over this range, four different failure models are identified, as illustrated in Fig. 7.

In mode (*i*), for smaller aspect ratio, the central part of the triangle is largely unyielded, with the plug falling vertically against a boundary layer along the left wall. As for a slender rectangle, a plastic region forms underneath, buffered from a stagnant lower left corner by a narrow shear layer. Unlike the rectangle, however, the central falling plug again yields near the top of the triangle, with the boundary layer peeling off the wall to divide another plastic zone from the plugged pinnacle.

Further details of mode (*i*), along with complementary slipline solutions are shown in Fig. 8. The lower section is similar to that arising in failure mode (*i*) of a rectangle, with the slanting of the side surface lengthening the stress discontinuity contained in the slipline field. The upper plastic zone has a simpler slipline pattern composed of a checkerboard of curved cycloidal sliplines extending from the side to a point along the left wall where two orthogonal slipline meet that boundary tangentially or normally.

The triangle's mode (*ii*) arises when the aspect ratio becomes sufficiently large to allow the falling plug to break free of the side wall and begin to rotate; see Fig. 7(b). Again the shear layer bordering to the left no longer attaches to the wall, but touches it tangentially. The transition from mode (*i*) to (*ii*) is thus analogous to that for the rectangle. However, for higher aspect ratio, the flow pattern for the triangle's mode (*ii*) becomes more complicated as the lower plastic region gradually plugs up, whereas the upper plastic zone largely disappears. Mode (*iii*) for the triangle arises when the rotation centre of the lower plug moves to the left of the lower right corner (Fig. 7(c)). Finally, mode (*iv*) arises near $X_0 = 1$ when the moving portions of the triangle largely plug up, to leave a nearly circular arc of failure and small plastic zones at the corners (Fig. 7(d)).

Diagnostics of the failure modes are shown in Fig. 9. Again, Od_c monotonically increases with X_0 over the range $0 < X_0 \leq 1$, with $\mathrm{Od}_c \approx X_0$ for $X_0 \ll 1$, as seen in Fig. 9(a). However, the reflection symmetry implies that this trend reverse for $X_0 > 1$. Fig. 9(b) shows the diagnostic $X_0 - x_b$, where x_b denote the position of the shear-rate maximum along the bottom wall. That position always lies at the corner $x = X_0$ for modes (*i*) – (*ii*). When the rotation centre of the lower plug moves to the left of the corner, however, the shear-rate maximum approximately locates its horizontal position, and so the plot in Fig. 9(b) exposes the transition to mode (*iii*). The third panel of the figure shows the position $z = z_b$ of the shear-rate maximum along the left wall, as well as the region over which $\dot{\gamma}$ reaches 10% of that maximum (indicated by the shading). For mode (*i*), z_b lies somewhere within the wall boundary layer and the shading roughly indicates its span. The transition from modes (*i*) to (*ii*) is identified by when the wall layer shortens to the thickness of the circular shear layer.

Fig. 9(c) also illustrates another key difference between mode (*i*) for rectangles and triangles: in the limit $X_0 \to 0$, the falling plug persists over the full length of the wall for the rectangle. By contrast, numerical solutions indicate that the falling plug in the triangle becomes gradually shorter in this limit, with the upper plastic zone migrating downwards. These trends are confirmed by the slipline solutions. In more detail, the slipline constructions predict that the vertical span of the descending plug becomes $O(X_0^{1/2})$ for $X_0 \to 0$. In other words, slender rectangular columns collapse fully by sliding along the side wall, whereas triangular blocks adhere to that wall and collapse only locally at their base, a feature that can be verified asymptotically in the slender limit Appendix D.

Finally, we may again construct lower bounds to the threshold Od_c using trial velocity fields. The purely rotating configuration illustrated by the insets of Fig. 9(a) proves more effective than a combined vertical-rotational configuration for the triangle. This lower bound construction leads to the predictions also included in Fig. 9 (see Appendix E for more details of the underlying calculations). Here, we also show the slipline predictions for mode (*i*). The circular arcs for the lower bound are added to Fig. 7, working surprisingly well for all but mode (*i*).

3.3. Drums

Of all our three shapes, the flow patterns in the drums are by far simplest, at least when the fill fraction f is not too small: when $f \gtrsim 9.08\%$, the fluid invariably fails over a narrow, viscously broadened, circular arc connecting the two most distant positions on the free surface; see Fig. 10. For lower fill fractions, the arc follows the lower surface of the drum (Fig. 10(a,b)). For higher fill fraction, the arc moves up into the fluid (Fig. 10(c–f)). The fill fraction at which the arc of failure leaves the drum surface is $f \approx 25.6\%$. In fact, once the failure arc ascends into the fluid, the flow patterns for all fill fraction are identical, but for very minor adjustments over the ends of the shear layers to accommodate the slightly different inclination of the drum walls. The critical threshold Od_c and the radius of the failure arc (obtained from a least squares fit of the plug velocity field) are plotted against f in Fig. 11 for a wider suite of numerical solutions.

Because the flow pattern is so simple, the perfectly plastic solution can be constructed analytically using the lower-bound calculation. For

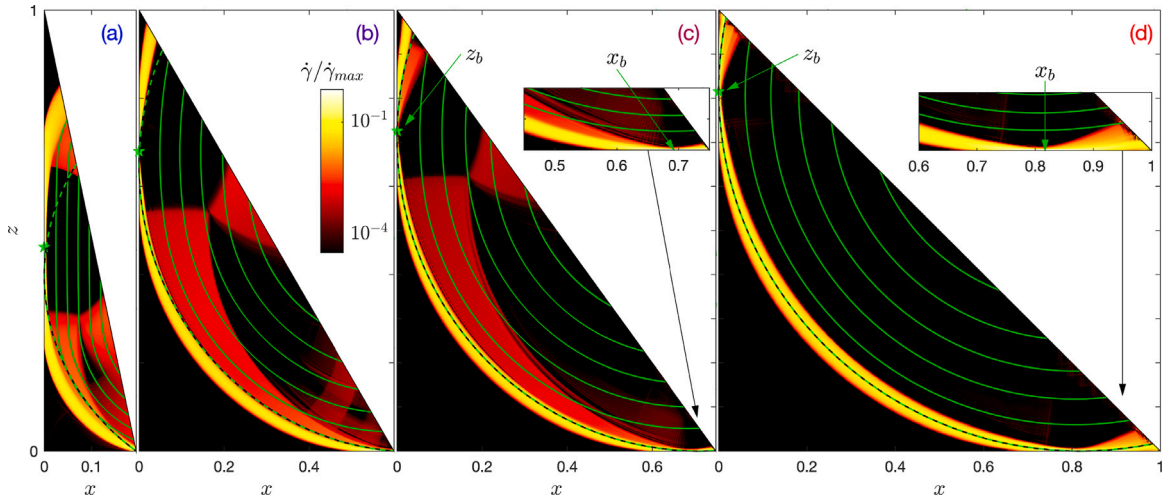


Fig. 7. Failure modes for triangles with (a) $X_0 = 0.2$, (b) $X_0 = 0.6$, (c) $X_0 = 0.75$, (d) $X_0 = 1$. Shown are selected streamlines superposed on logarithmic density maps of strain rate $\dot\gamma$. The insets of (c,d) show magnifications near the bottom right corner. The locations of maximum shear rate along the bottom and left walls, x_b and z_b, are indicated. The (green and black) dashed lines show the circular failure arcs of the corresponding rotational configurations of the lower bound (see Fig. 9(a)).

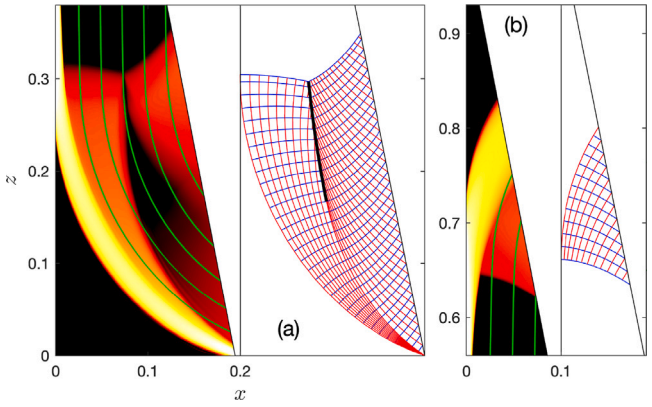


Fig. 8. Further details of failure mode (i) for a triangle with $X_0 = \frac{1}{5}$, showing selected streamlines superposed on a logarithmic strain rate density plot on the left, and the corresponding slipline construction on the right. The lower plastic zone is presented in (a); the upper one in (b). The thicker solid line indicates the stress discontinuity predicted by slipline theory.

this task, we adopt trial velocity fields with a single circular failure arc and a rotation centre controlled by the optimization angle φ shown in Fig. 10. Analytical details of the construction are provided in Appendix F. When $9.08\% \lesssim f \lesssim 25.6\%$, the rotation centre lies at the drum centre, and the circular failure arc lies along the drum surface. In this case, $f = (\varphi - \sin\varphi)/(2\pi)$ and $\mathrm{Od}_c = \sin^2\frac{1}{2}\varphi/(3\varphi)$. For $f \gtrsim 25.6\%$, the failure arc pierces the fluid, with $\varphi \approx 2.33$ and $\mathrm{Od}_c \approx 0.0854$ (this value of φ is the solution to $\varphi = \tan\frac{1}{2}\varphi$). The resulting predictions for the rotation arc are added to Fig. 10. Corresponding results for critical thresholds Od_c and the radius of the failure arc are included in Fig. 11.

For very small fill fraction, $f \lesssim 9.08\%$, the failure mode becomes different, as illustrated in Fig. 12. In such relatively slender configurations, the rotating plug no longer remains intact, but breaks over its thinnest sections. This breakage introduces plastic zones near the edges of the segment that widen relative to the length of the free surface as f continues to decrease down to 1%. These zones are again characterized by cycloidal sliplines, as shown by the inset of Fig. 12. For lower fill fraction still, the plastic regions migrate to the centre of the segment,

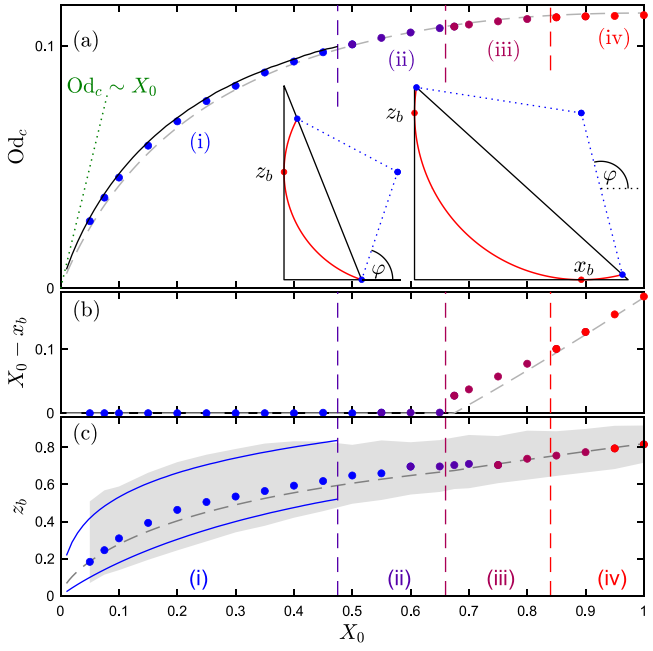


Fig. 9. Diagnostics of failure modes for triangles: (a) Od_c, (b) the distance of the shear rate maximum along $z = 0$ from the lower right corner, $X_0 - X_b$ and (c) the location of the shear-rate maximum along the left-hand wall, z_b. The dots show numerical results, the solid line indicate the narrow-width slipline calculation, and dashed lines show lower bounds (for a single rotational configuration). The plots are divided into the four modes identified, and colour-coded accordingly. In (a), the threshold in the slender limit, $\mathrm{Od}_c \sim X_0$, is indicated, and the insets show the rotational configurations used for the lower bound, along with the optimization angle φ. In (c), the shaded region locates the boundary layer along $x = 0$ (shading grey the locations where $\dot\gamma$ is above 1/10th of its maximum value). The two solid lines indicate the corresponding slipline predictions for the top and bottom of that boundary layer.

leaving behind rigid plugs attached to the drum at the edges, and the resulting failure mode develops similarities to the mode (*i*) of a slender triangle.

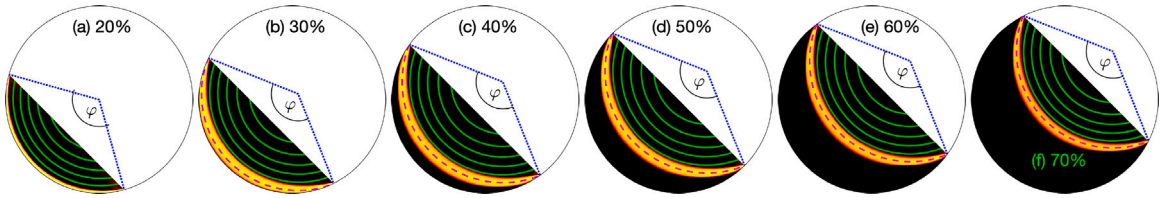


Fig. 10. Failure modes of drums for varying fill fraction, showing selected streamlines superposed on a logarithmic strain rate density plot. The angle subtended by the rotating plug at its rotation centre is φ and corresponds to the optimization parameter of the lower bound. The dashed lines show the result for the failure arc from the lower-bound calculation; blue dotted lines indicate the associated failure-arc radius r_p.

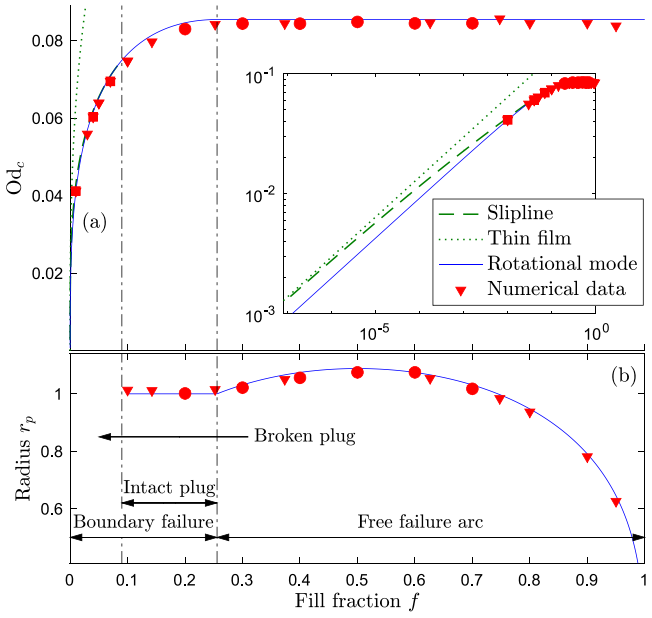


Fig. 11. (a) Critical threshold Od_c against fill fraction f for the drum. (b) Corresponding radius r_p of the failure arc for the cases with an unbroken rotating plug, scaled by drum radius. The vertical dot-dashed lines indicate the fill fractions at which either Od_c becomes constant or the plug breaks. The filled circles and squares indicate numerical results from the solutions displayed in Figs. 10 and 12, respectively; the triangles show additional numerical results. The solid lines show the predictions assuming rotational failure mode along a circular arc (see (F.7), (F.4) and (F.6) in Appendix F). In (a), the dotted line shows the thin-film result in (3.1), and the dashed line is the prediction from a slipline construction similar to that for mode (i) for a triangle Appendix C. The inset replots the data using logarithmic scales.

Eventually, for $f \to 0$, only the central part of the segment yields and the threshold for motion is expected to become

$$\mathrm{Od}_c \sim \frac{\sqrt{2}}{16}\,(12\pi f)^{1/3}\,, \tag{3.1}$$

which follows from thin-film theory (see Appendix D). In Fig. 11, it is hard to detect the switch over to this limit from the prediction for Od_c for the simple rotational failure mode Appendix F, which still tracks the numerical results for the fill fractions shown. However, in Fig. 11 we further include the prediction from a slipline construction, which largely follows the same analysis for mode (i) of a triangle Appendix C: cycloidal plastic zones are constructed at geometrically controlled positions on either side of the segment. The critical threshold Od_c then follows from demanding torque balance on the remnant of the rotating plug. As shown by the inset to Fig. 11(a), the slipline construction captures the (rather protracted) switch over from the rotational failure mode for $f \gtrsim 9.08\%$ to the thin-film result in (3.1) for $f \to 0$.

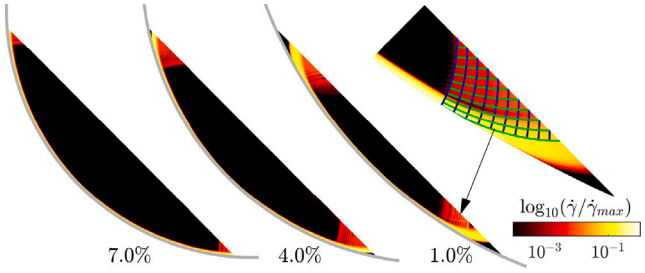


Fig. 12. Failure modes for drums with low fill fraction ($f = 7\%, 4\%, 1\%$). Grey lines indicate the drum boundary. For fill fraction $f = 1\%$, the inset shows a magnification of the plastic region at lower end of the segment, with constructed cycloidal sliplines.

To determine the fill fraction at which the rotating plug first appears on raising f (or, equivalently, when that plug first breaks on lowering f), we note that the plastic zones for smaller f feature sliplines that leave the drum surface tangentially before meeting the free surface at 45°. This becomes impossible at the corners of the segment when the slope of the drum surface relative to the free surface also reaches 45°. In other words, when the drum surface becomes vertical or horizontal at the corners of the segment and $f = (\pi - 2)/(4\pi) \approx 9.08\%$, the plastic zones must disappear there.

4. Discussion

In this paper, we have derived the modes of failure of three different shapes for a two-dimensional block of viscoplastic material. The failure modes correspond to the flow patterns that arise at the onset of motion, and are the solutions of a nonlinear eigenvalue problem in which the threshold for motion plays the role of the eigenvalue. In this sense, the failure modes are the counterparts of the most dangerous modes of the linear stability analysis that applies for a simpler medium with a linear constitutive law. Here, the threshold arises as the critical value of the yield stress in comparison a characteristic gravitational stress.

Our calculations are partly numerical and have been performed using the Bingham constitutive law. However, because the failure mode applies to the perfectly plastic limit of the problem, the fact that this law has constant (plastic) viscosity becomes irrelevant, and our results apply more generally. Indeed, we have built analytical solutions to complement numerical results for this problem using slipline and limit analyses from plasticity theory, and so the only important detail of the constitutive law in the plastic limit is that it follows the von Mises yield condition.

The failure modes of a rectangle or triangle are functions of aspect ratio, and take the form of a patchwork pattern that combines plugs with viscoplastic boundary layers and zones of perfectly plastic deformation. The drum is simpler when fill fraction is not too low. The failure mode always takes the form of a circular failure arc that divides an overlying rotating plug from either the drum surface or an

underlying stationary plug. For all three cases, we have characterized the failure modes by their thresholds and other geometrical properties.

To compute the failure modes, we use a limiting procedure in which solutions are constructed that are beyond the threshold for motion and already yielded. As one reduces the dimensionless yield stress towards the threshold, the velocity field decreases, until it vanishes at onset. Our solutions therefore correspond to snapshots of an initial condition. Indeed, for any finite time beyond this threshold, the motion of the fluid changes the geometry, which in turn adjusts the stress and velocity field. We have not tracked how that evolution impacts the flow patterns, which would require the use of a general numerical solver of the full initial-value problem with an embedded interface tracking scheme.

The restriction to the initial snapshot also implies that our results carry over to some other related problems. For example, as demonstrated in Appendix B, the solutions also apply to scenarios in which the vertical axis is reflected. In other words, instead of a block that collapses under vertical compression due to its own weight, the failure modes also apply to the opposite problem in which a hanging block fails in vertical extension; that is, to dripping-type problems [20–22] or the viscoplastic Rayleigh–Taylor instability [23,24]. Admittedly, for a rectangle or triangle in that setting, the free-slip constructions described in [11] would be more appropriate, given that such side boundary conditions apply along the centreline of a hanging thread. Importantly, at any finite time after flow is initiated, the resulting changes in geometry distinguish failure under compression from that in extension. In particular, yielded drips accelerate because the thinning of the fluid weakens the structure, whereas collapses under compression eventually widen the fluid's aspect ratio, strengthening the configuration and decelerating flow. For drums, the gravity-inverted versions of our results imply that yield-stress fluids clinging to the underside of the drum actually rotate out of position when yielding, rather than flowing mostly vertically, as might perhaps be expected.

Finally, one wonders whether failure modes of the type constructed here could be visualized experimentally. The drum configuration, in particular, offers a convenient setting for a laboratory experiment, given that fluid can be easily loaded into a section cut from a cylinder and the surface levelled. Placing the cylindrical section onto rollers that can be gradually rotated then offers the possibility of slowly steepening the surface towards failure. Provided the fill fraction is not too low, one then expects a top portion of the fluid to rotate out of position, leaving behind fluid adhering to the drum with a cylindrical surface of a different radius. Based on our calculations, one expects such motion to arise when surface angle α exceeds a critical value α_c dependent on the fill fraction of the drum f, the length of the free surface $\mathcal{L}(f)$, and the fluid yield stress and density, τ_Y and ρ (respectively). Given the function $\mathrm{Od}_c(f)$ displayed in Fig. 11, and restricting attention initially to $\alpha < \frac{1}{2}\pi$, the material should mobilize when

$$\sin\alpha > \sin\alpha_c, \quad \alpha_c = \sin^{-1}\left\{\mathrm{Min}\left[1, \frac{\tau_Y\sqrt{2}}{\rho g \mathcal{L}(f)\mathrm{Od}_c(f)}\right]\right\}, \tag{4.1}$$

where g is gravity.[1]

The inequality in (4.1), follows from the condition $\mathrm{Od} < \mathrm{Od}_c(f)$ and our definition of the dimensionless yield stress Od. In fact, it applies equally to inclination angles $\alpha > \frac{1}{2}\pi$, and demands that failure must also arise when $\alpha < \pi - \alpha_c$. This result again follows from the vertical reflection symmetry, which states that the failure criterion is the same whether the fluid fills the bottom portion of the drum or clings to the upper surface. Overall, then, failure arises for angles over the range $\alpha_c < \alpha < \pi - \alpha_c$, as illustrated in Fig. 13 for different values of the parameter $\tau_Y/(\rho g R)$ (an Oldroyd number based on the radius of the drum). At sufficiently low or high fill fraction, the critical angle always reaches 90°; the plot shows the true symmetry about $\alpha = 90°$, as well as an approximate reflection symmetry about $f = \frac{1}{2}$. The limiting value 0.0854 reached by $\mathrm{Od}_c(f)$ for $f \gtrsim 0.265$ translates to a critical value $\tau_Y/(\rho g R) \approx 0.08754\sqrt{2} \approx 0.1208$, above which there can be no failure whatever the surface inclination.

[1] The length of the free surface $\mathcal{L}$ is equal to $2R\sin\frac{1}{2}\beta$, where R is the drum radius, $f = (\beta - \sin\beta)/(2\pi)$, and β is the angular coverage of the drum surface by the fluid, as indicated in Fig. 1.

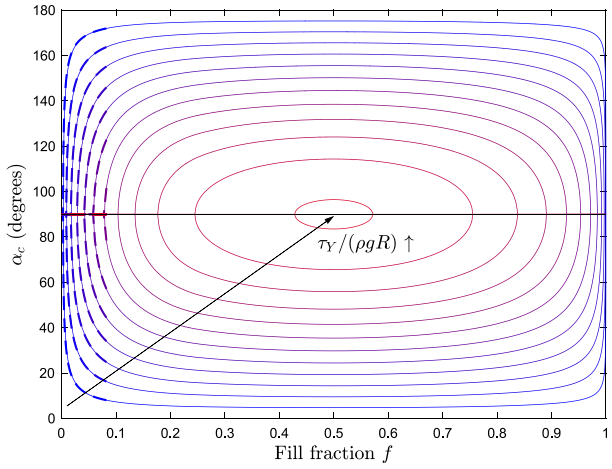


Fig. 13. The critical failure angle α_c for drums with fill fraction f and different values of the quantity $\tau_Y/(\rho g R)$ (from 0.01 to 0.12 in steps of 0.01, colour coded from blue to red); failure occurs inside each of the contours. The solid lines show the predictions of the rotational failure mode; these are corrected using the slipline solution at small f (dashed lines), although the difference is hardly visible.

Of course, our entire discussion has focused on two-dimensional problems, whereas any experimental realization or application must be three-dimensional. Awkwardly, the methodology that we have used here does not easily extend beyond two dimensions (both our slipline theory and numerical method are restricted to two-dimensional Cartesian coordinates). We must, therefore, leave consideration of the three-dimensional problem to future work. Similarly, we have also ignored more complicated rheological effects, such as elasto-viscoplasticity. Elasticity allows for the finite deformation of the material below yield stress, which introduces the notable complication that the failure modes must characterize shape that is already deformed. Dealing with this feature requires a time-dependent elasto-viscoplastic constitutive law, another extension we leave for future work.

CRediT authorship contribution statement

J. Dai: Writing – original draft, Investigation, Formal analysis. **N.J. Balmforth:** Writing – review & editing, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **D.R. Hewitt:** Writing – review & editing, Software, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Numerical implementation strategy

To solve the equations in (2.17) to (2.19), we employ an augmented Lagrangian algorithm [25,26]. More details of the specific version of this algorithm that we use can be found in [18]. Importantly, the implementation exploits a Fourier-series description of the solutions

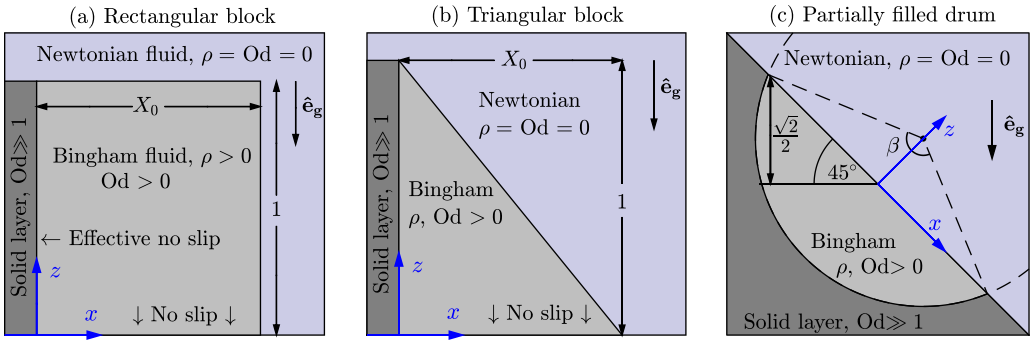


Fig. A.14. Dimensionless version of the sketch of Fig. 1, including the buffer layers introduced to enforce no-slip, for (a) a rectangular block, (b) a triangle, and (c) a partially filled drum with angular coverage $\beta = \frac{3\pi}{4}$ and fill fraction of $f \approx 26.2\%$. In each case, the computational domain is shown and the darker grey regions indicate the buffer layers. The lighter blue region shows the area over which we emplace an ambient viscous fluid with the same viscosity as the Bingham fluid (shaded in light grey) to provide a stress-free condition in the plastic limit.

in one of the space dimensions; here, that direction is taken to be x. Consequently, the representation demands the imposition of symmetry conditions on the two sides of the computational domain.

To enforce the no-slip condition, we therefore include a buffer layer adjacent to the boundary over which the yield stress is increased by fifteen orders of magnitude (i.e. the material is effectively enforced to be rigid) and the density is the same as the Bingham fluid (see Fig. A.14). For rectangles and triangles, the buffer layers have a dimensionless thickness of 0.05 and a height of unity. For the drum, we use a square computational domain and the precise details depend on the fill fraction of the drum. When the fill fraction is below 50%, we place the drum's free surface at the centre of the domain, and set the domain's diagonal to be 1.4 times the free surface's length. When the fill fraction is larger than 50%, we instead place the drum's centre at the domain centre to include the entire drum within the domain. The domain size is then chosen to be 1.4 times the drum diameter. In both cases, the buffer corresponds to the region outside the drum over the half-plane below the inclined free surface, as sketched in Fig. A.14(c).

The boundary conditions along the free surfaces are dealt with in a different way: we employ a volume-of-fluid approach in which we further supplement the fluid domain with additional buffer zones above and to the right. Over these buffers, we include a second fluid with zero density and yield stress, but the same viscosity as our Bingham fluid. In other words, we turn the problem into a two-phase one. Then, in the plastic limit in which the viscous stresses become insignificant over the buffer zones, the continuity of the stress translates to a free-surface condition. This is the same technique that was used in some of the computations in [11]. The resulting geometry of the computational domain is illustrated in Fig. A.14 for all three geometries we consider for the failing block.

With the augmented-Lagrangian algorithm in hand, which returns the solution for a given value of Od, we construct solutions that approach the plastic limit as follows. We begin at a low value of Od, and compute the solution (resetting Od if the chosen value turns out to be too large, such that there is no flow). We then incrementally increase Od, monitoring the maximum speed achieved in the computation over the domain of the Bingham fluid to assess the proximity of the threshold Od_c (where that speed must vanish). Having constructed a sequence of solutions with maximum flow speeds, and reached a pre-determined, small threshold for the maximum speed,[2] we terminate computations, take the nearest solution to the plastic limit to approximate the failure mode, and linearly extrapolate to estimate Od_c. Sample sequences of solutions constructed in this manner are illustrated in Fig. A.15.

[2] 4×10^{-5} for the rectangle, 3×10^{-4} for the triangle, and 5×10^{-4} for the drum.

For most of the computations, the spatial resolution of the numerical grid was 1024×1000 for rectangles and triangles, and 512×512 for the drum. For the calculations shown in the paper's figures, we doubled this spatial resolution in order to confirm that the solutions were adequately resolved. At each calculation of fixed geometry and fixed Od, the iteration stops when the relative strain rate error between two consecutive iterations inside the Bingham fluid is smaller than 10^{-3}. To ensure the Newtonian ambient does not effect the Bingham fluid, we assigned 30% of horizontal grids and 15% of vertical grids to the ambient fluid in the cases of rectangles and triangles. We verified that such solutions were insensitive to our methodology for implementing the boundary conditions.

Appendix B. Symmetries

B.1. Triangles

Consider a solution to Eqs. (2.17)–(2.19) for a triangle with base width X_0. Let

$$\hat{x} = \frac{z}{X_0}, \quad \hat{z} = \frac{x}{X_0}, \quad \hat{u} = -\frac{w}{X_0}, \quad \hat{w} = -\frac{u}{X_0}. \tag{B.1}$$

The transformed strain rate tensor has independent components, $\hat{\dot{\gamma}}_{\hat{x}\hat{x}} = \dot{\gamma}_{xx}$ and $\hat{\dot{\gamma}}_{\hat{x}\hat{z}} = -\dot{\gamma}_{xz}$, whereas the Bingham constitutive law (2.19) demands $\hat{\tau}_{\hat{x}\hat{x}} = \tau_{xx}$, $\hat{\tau}_{\hat{x}\hat{z}} = -\tau_{xz}$. If we define

$$\hat{p} = \rho g(\hat{z} + X_0\hat{x} - 1) - p, \tag{B.2}$$

then Eqs. (2.17)–(2.19) all remain unchanged, but for the introduction of the new variables. For the boundary conditions, the no-slip conditions remain satisfied as the transformation simply swaps the bottom and left wall. The free surface conditions become

$$\hat{\boldsymbol{\tau}} \cdot \begin{pmatrix} X_0 \\ 1 \end{pmatrix} = \mathbf{0}, \quad \text{at } \hat{z} + X_0\hat{x} = 1, \quad 0 \le \hat{x} \le X_0^{-1}. \tag{B.3}$$

In other words, the original problem for a triangle of base width X_0 is equivalent to that for a triangle with base with X_0^{-1}. This symmetry is illustrated for two numerical solutions in Fig. B.16. Note, however, that this symmetry is valid only at initial state, and is immediately lost at any finite time once the fluid flows.

B.2. Vertical reflection

When the geometry is reflected about the x-axis, the original failure problem becomes equivalent to a dripping or Rayleigh–Taylor problem. We introduce the transformation.

$$(\bar{x}, \bar{z}) = (x, -z), \quad (\bar{u}, \bar{w}, \bar{p}) = (-u, w, -p),$$

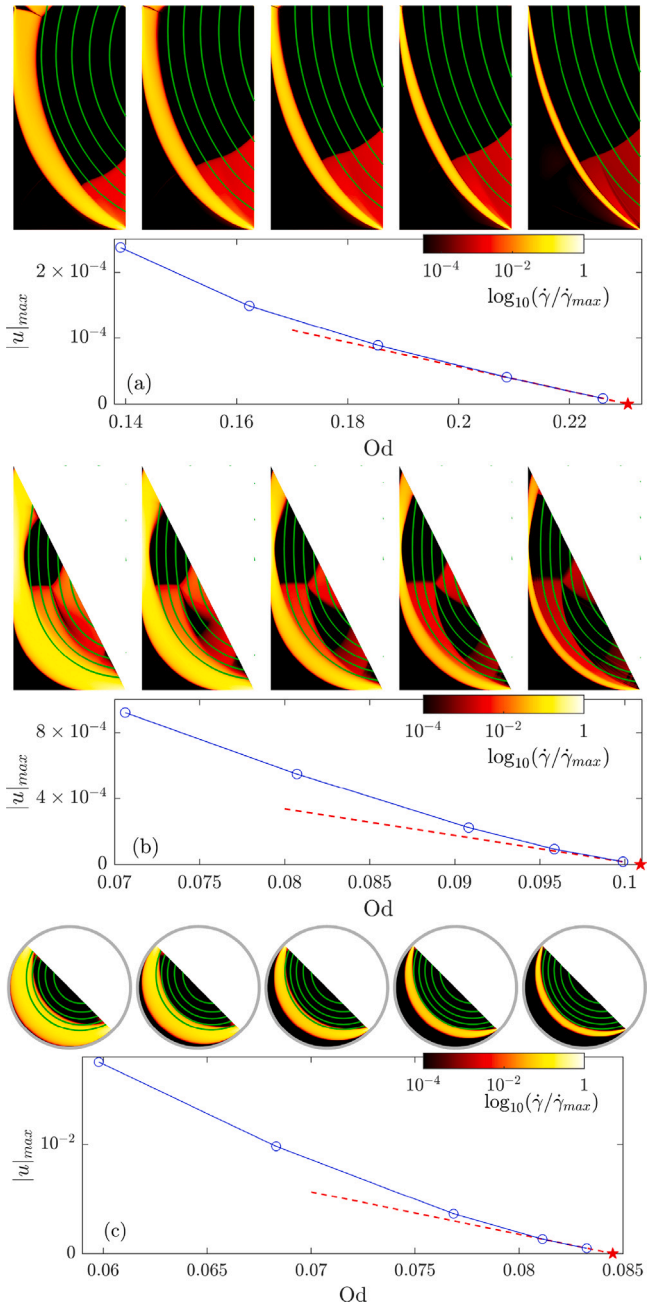


Fig. A.15. A sequence of solutions for (a) a rectangle width $X_0 = \frac{1}{2}$, (b) a triangle with base width $X_0 = \frac{1}{2}$, and (c) a drum with a fill fraction of 50%. For each shape, the top row presents five solutions with increasing Od, showing sample streamlines superposed on logarithmic density maps of $\dot\gamma/\dot\gamma_{max}$. In (c), the grey circle indicates the drum surface. The ambient fluid region is coloured white. In the lower panels, we plot the maximum speed $|\mathbf{u}|_{max}$ over the domain of the Bingham fluid against Od. The circles indicate the solutions appearing the upper rows and the red dashed line and star indicate the final linear extrapolation to Od_c.

to find the same model equations, but with the no-slip boundary on the top and the free-slip surface underneath. In other words, the collapse of a tall, thin column is equivalent to the dripping of a slender thread; the collapse of a wide block corresponds to the Rayleigh–Taylor failure of a slab on the underside of a horizontal plate. Again, the symmetry applies only to the state at initial failure, with any subsequent flow immediately distinguishing each problem.

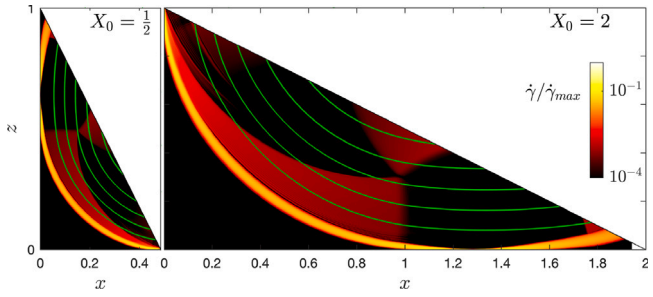


Fig. B.16. Two numerical solutions, one with a base width of a half, the other with a base width of two, showing the symmetry of the strain rate (logarithmic density plot) and velocity (streamlines) under a specific reflection and scaling.

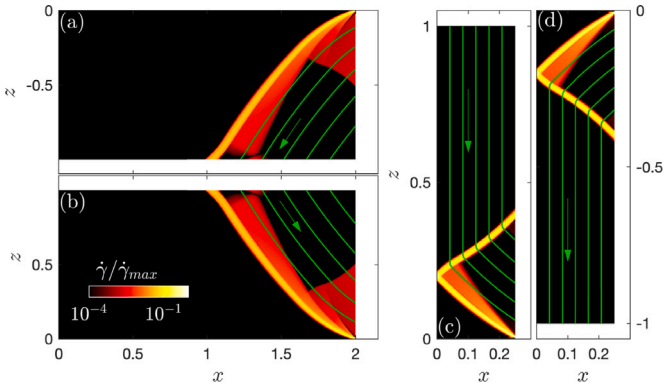


Fig. B.17. Numerical solutions for rectangular blocks illustrating the vertical reflection symmetry. In panels (a) and (b), we show solutions for rectangles with $X_0 = 2$, using the vertically reflected or original geometry, respectively. Panels (c) and (d) show a similar pair of solutions for rectangles with $X_0 = \frac{1}{4}$. In this latter pair, a symmetry (or free-slip) condition is imposed at $x = 0$, as in [11]. Shown are logarithmic density plots of the relative strain rate, with a selection of streamlines overlaid. The arrows indicate the flow direction.

We illustrate the vertical reflection symmetry in Fig. B.17. Panels (a) and (b) show a relatively wide rectangular block, whilst panels (c) and (d) display a more slender configuration. To make the latter more similar to a dripping problem, we depart from the set-up used elsewhere in this paper and impose a symmetry (or free-slip) condition along the left wall in this pair of examples. No-slip is imposed for the first pair of solutions, although the results would be identical with free-slip because this boundary is cloaked by the adjacent plug.

The wide rectangular block in Fig. B.17(a) corresponds to the set up of a Rayleigh–Taylor instability in which a slab underneath a horizontal plate fails near the free side surface. Had the slab been fully contained between vertical side walls, the Rayleigh–Taylor instability would have been completely eliminated [23,24] (the gravitational stress being accommodated within the hydrostatic pressure, and therefore not impacting the yield condition). The free surface at the side, however, permits the lower right corner to fail. The breakage of the slender thread in Fig. B.17(d) displays similarities with the necking observed in viscoplastic dripping (although the geometry there is axisymmetric) [20–22].

Appendix C. Slip-line constructions

Following [10,11], the construction of the sliplines begins from the free side surface. For plastic regions under compression, the boundary conditions require that

$$P = \mathrm{Od}(z+1), \quad \theta = \frac{3\pi}{4} + \chi \quad \text{at} \quad x = X_0 - z\tan\chi, \tag{C.1}$$

where $\chi = 0$ for rectangles and $\chi = \tan^{-1}(X_0)$ for triangles. The resulting checkerboard of curved sliplines are actually cycloids. This follows because the boundary conditions on the side surface coincide with the symmetry conditions applying along the midline of a plastic layer sandwiched between two plates. Therefore, Prandtl's classical cycloid solutions for that problem also apply here [6,7,18].

More explicitly, the cycloids for the plastic region in compression can be parameterized as follows:

$$\left.\begin{aligned} x &= x_0 + \mathrm{Od}\sec\chi\cos t_\alpha, \\ z &= z_0 + \mathrm{Od}\sec\chi(1 - \tfrac{1}{2}\pi + t_\alpha - \sin t_\alpha), \end{aligned}\right\} \quad \alpha-\text{lines}, \tag{C.2}$$

and

$$\left.\begin{aligned} x &= x_0 + \mathrm{Od}\sec\chi\cos t_\beta, \\ z &= z_0 + \mathrm{Od}\sec\chi(1 + \tfrac{1}{2}\pi - t_\beta - \sin t_\beta), \end{aligned}\right\} \quad \beta-\text{lines}. \tag{C.3}$$

Here (x_0, z_0) denotes a point on the side surface. Below the web of cycloids, the characteristics are composed of the fan from the bottom right corner. Here, the slipline angles begin over the range, $\frac{3}{4}\pi < \theta < \theta_a$, for some final angle θ_a. The full slipline field can be constructed by continuing the cycloids through the fan using standard finite-difference methods from plasticity theory [7].

C.1. Rectangles

For the rectangle, the plastic regions are bounded by a lowermost α-line and an uppermost β-line. For mode (i), the plastic region ends at the left no-slip wall, where the slipline angle must satisfy $\theta = \frac{1}{2}\pi$. The critical threshold Od_c and θ_a then follow from ensuring that the last slipline in the fan, with $\theta = \theta_a$ at $(x, z) = (X_0, 0)$, reaches $x = 0$ with $\theta = \frac{1}{2}\pi$, and ensuring forces are balanced on the descending plug. The latter amounts to computing the net vertical force along the top $\beta-$line and equating this to the gravitational force on the plug, less the resistance from the boundary layer along the side wall (which is Od times its length). The calculation is eased by first scaling the problem slightly differently, thereby avoiding any need for an iterative procedure; *cf.* [10].

For mode (iii), where the plug is rotating, the plastic region ends within the block, with a slipline angle of θ_b. The leftmost $\alpha-$line then continues upwards as a circular arc that ends at the left top corner. At the connection to the circular arc, the slipline angle θ remains continuous, but the curvature does not. Both θ_b and the slipline angle at the top left corner, θ_c, become parameters of the construction. The critical threshold Od_c and angles $\{\theta_a, \theta_b, \theta_c\}$ now follow from the continuity of slipline angle at the end of the fan, and from balancing the net torque and horizontal and vertical force on the rotating plug.

Note that when the slipline field terminates to the left along an $\alpha-$line that steepens too close to $\theta = \frac{1}{2}\pi$, a stress discontinuity appears that disfigures the pattern. The discontinuity is always present for mode (i) in view of the boundary condition at $x = 0$, and may be present in mode (iii). The feature can be dealt with by applying the jump conditions [10],

$$P^+ - P^- = 2\mathrm{Od}\sin(\theta^+ - \theta^-), \tag{C.4}$$

$$\gamma = \frac{1}{2}(\theta^+ + \theta^-), \tag{C.5}$$

and constructing the locus of the discontinuity along with the sliplines. Here, the $\pm$ superscripts refer to the two sides of the discontinuity, and γ is the slope of its locus.

C.2. Triangles

For mode (i) in triangles, the construction is similar to that outlined above for rectangles, including the need for the inclusion of a stress discontinuity below the falling plug. However, a second plastic region lies above that plug. This plastic region is in tension, and the side boundary condition there is

$$P = \mathrm{Od}(z - 1), \quad \theta = \frac{\pi}{4} + \chi. \tag{C.6}$$

The sliplines are again cycloids, but with slightly different parameterization and range:

$$\left.\begin{aligned} x &= x_0 + \mathrm{Od}\sec\chi\cos t_\alpha, \\ z &= z_0 - \mathrm{Od}\sec\chi(1 - \tfrac{1}{2}\pi + t_\alpha - \sin t_\alpha), \\ \tfrac{1}{2}\pi &\le t_\alpha \le \tfrac{1}{2}\pi + \tfrac{1}{2}\mathrm{Od}^{-1}(z_0 - z_H), \end{aligned}\right\} \quad \alpha-\text{lines}, \tag{C.7}$$

$$\left.\begin{aligned} x &= x_0 + \mathrm{Od}\sec\chi\cos t_\beta, \\ z &= z_0 - \mathrm{Od}\sec\chi(1 + \tfrac{1}{2}\pi - t_\beta - \sin t_\beta), \\ \tfrac{1}{2}\pi &\le t_\beta \le \pi - 2\chi - \tfrac{1}{2}\mathrm{Od}^{-1}(z_0 - z_H), \end{aligned}\right\} \quad \beta-\text{lines}. \tag{C.8}$$

Here, (x_0, z_0) is again a point on the free surface, with $z_H \le z_0 \le z_H + \mathrm{Od}(\pi - 4\chi)$. Here z_H is the lowest height of the upper plastic region at the free surface. The boundary condition demanding that the lowest slipline reach the no-slip wall at $\theta = \frac{1}{2}\pi$ still holds, and is satisfied by one of the cycloids, so an expansion fan is not needed.

The falling plug is now bounded by two sliplines, a piece of free surface, and a boundary layer at the left wall. Finally, the three parameters θ_a, z_H and Od_c are determined by the boundary conditions $\theta = \frac{1}{2}\pi$ at $x = 0$ and the vertical force balance on the falling plug. This analysis can be adapted to construct slipline solutions for slender configurations in a drum with low fill fraction.

Appendix D. Slender configurations

In the slender limit, $X_0 \ll 1$, failure must arise mostly by fluid sliding against the left wall for both a rectangle and triangle. In this shear-dominated state, the $x-$derivatives must be $O(X_0^{-1})$ larger than the vertical derivatives, and the incompressibility condition demands that $u = O(X_0 w)$. Where the fluid is yielded, it follows that $\tau_{xx} = O(X_0\tau_{xz})$. The vertical force balance then implies

$$\frac{\partial\tau_{xz}}{\partial x} \sim -1, \tag{D.1}$$

given that the stress conditions on the right side imply that

$$\tau_{xz} = \frac{2X_0\tau_{xx}}{1 - X_0^2} = O(X_0^2\tau_{xz}), \tag{D.2}$$

$$p = \frac{1 + X_0^2}{1 - X_0^2}\tau_{xx} = O(\tau_{xx}), \tag{D.3}$$

which, in conjunction with the horizontal force balance (involving all three terms), demands that the pressure must be $O(\tau_{xx})$ everywhere fluid is yielded. Hence,

$$\tau_{xz} \sim x - X_0 + z\tan\chi, \tag{D.4}$$

where χ is 0 for the rectangle and $\tan^{-1}\chi$ for the triangle (as in Appendix C).

The fact that $\tau_{xx} = O(X_0\tau_{xz})$ implies that the yield condition reduces to

$$\tau \sim |\tau_{xz}| \sim |x - X_0 + z\tan\chi| > \mathrm{Od}, \tag{D.5}$$

in the slender limit. For the rectangle, the yield condition is therefore first met at $x = 0$ when $\mathrm{Od} = \mathrm{Od}_c = X_0$. For the triangle, the corresponding condition yields the same threshold, but fluid first yields only at $x = z = 0$.

For a drum, the analysis is similar, although the 45° inclination of the surface implies that the shear stress is now given by

$$\tau_{xz} \sim -\frac{z}{\sqrt{2}} \tag{D.6}$$

(given that the coordinates are now aligned with the free surface and the origin lies halfway along it). The shear stress is largest at the base

of the thickest part of the layer, which is where yielding must first take place, so that

$$\mathrm{Od}_c \sim \frac{h_{max}}{2\sqrt{2}}, \tag{D.7}$$

where the maximum depth $h_{max} \sim \frac{1}{4}(12\pi f)^{1/3}$. Hence

$$\mathrm{Od}_c \sim \frac{\sqrt{2}}{16}(12\pi f)^{1/3}. \tag{D.8}$$

Note that the failure mode computed in Appendix F is inconsistent with this result, as it implies a threshold of $\mathrm{Od}_c \sim \frac{1}{3}\sqrt{2}(6\pi f)^{1/3}$ as $f \to 0$ (see (F.7) and (F.4)). Thus, a simple rotational mode of failure with an arc aligned along the drum surface cannot apply for low fill fractions.

Appendix E. Rotational failure mode of a triangle

As indicated in Fig. 9, there are two cases to consider, depending on whether φ is less or greater than $\frac{1}{2}\pi$. For $\varphi < \frac{1}{2}\pi$, the failure arc grazes the side wall and the position of the centre of rotation is given by

$$(x_b, z_b) = r_p(1, \sin\varphi) \tag{E.1}$$

where the radius of the failure arc is given by

$$r_p = \frac{X_0}{1-\cos\varphi}. \tag{E.2}$$

The arc intersects both the bottom corner of the triangle and a position on the slanted side surface. The latter defines an (anti-clockwise and negative) angle χ with respect to the horizontal and centre of rotation given by $\chi = 2\tan^{-1} X_0 - \varphi$.

When, instead, $\varphi > \frac{1}{2}\pi$, the failure arc grazes the bottom wall and the position of the centre of rotation is given by $(x_b, z_b) = r_p(1, 1)$, with

$$r_p = \frac{X_0}{1-\cos\varphi} + X_0(1-\sin\varphi). \tag{E.3}$$

Once more, however, the angle fixing the upper intersection of the failure arc with the slanted side surface is $\chi = 2\tan^{-1} X_0 - \varphi$.

In the plastic limit, the (dimensionless) dissipation rate incurred by sliding along the failure arc with rotation rate Ω is

$$r_p^2\Omega(\varphi-\chi)\mathrm{Od}, \tag{E.4}$$

whereas the gravitational power released is

$$\frac{2}{3}r_p^3\Omega\sin^3\tfrac{1}{2}(\varphi-\chi)\cos\tfrac{1}{2}(\varphi+\chi). \tag{E.5}$$

Hence

$$\mathrm{Od} = \frac{2r_p\sin^3\frac{1}{2}(\varphi-\chi)\cos\frac{1}{2}(\varphi+\chi)}{3(\varphi-\chi)}. \tag{E.6}$$

Finding the maximum of this expression over φ gives the bound.

Appendix F. The failure mode of the drum

For the circular failure arc shown in Fig. 10 a plug rotates rigidly out of position. The plug has the form of a segment of a circle of (dimensional) radius Rr_p, and the arc has an angle φ. In the plastic limit, the dimensional dissipation rate arising along the failure arc is given by $\tau_Y R^2 r_p^2 \Omega\varphi$, where Ω is the rotation rate. Given that the area of the segment is $\frac{1}{2}R^2r_p^2(\varphi-\sin\varphi)$ and its centroid lies a distance $4Rr_p\sin^3(\frac{1}{2}\varphi)/[3(\varphi-\sin\varphi)]$ from the centre of rotation, the power input by gravity is

$$\frac{\sqrt{2}}{3}\rho g\Omega R^3 r_p^3\sin^3(\tfrac{1}{2}\varphi) \tag{F.1}$$

The power balance therefore implies that

$$\tau_Y r_p^2\varphi = \frac{\sqrt{2}}{3}\rho g R r_p^3\sin^3(\tfrac{1}{2}\varphi). \tag{F.2}$$

Or,

$$\mathrm{Od}_c = \frac{\tau_Y}{\rho g\mathcal{H}} = \frac{\tau_Y}{\sqrt{2}\rho g R r_p\sin(\varphi/2)} = \frac{\sin^2(\varphi/2)}{3\sqrt{2}\varphi}. \tag{F.3}$$

Optimizing this expression over φ implies that $\varphi = \tan(\varphi/2)$. Hence $\varphi \approx 133.6°$ and $\mathrm{Od}_c \approx 0.0854$. This optimal value only applies when the circular failure arc lies inside the drum. For fill fractions that are too low, however, the arc predicted for this value of φ lies outside the drum, in which case we must instead align the failure arc with the drum surface.

To determine the fill fractions f for which the failure arc aligns with the drum, we first note that

$$f = \frac{\beta-\sin\beta}{2\pi}, \tag{F.4}$$

where β denotes the angle subtended at the centre of the drum by the segment filled with Bingham fluid (*cf.* Fig. 1). The length of the free surface is then

$$2R\sin\tfrac{1}{2}\beta \equiv 2Rr_p\sin\tfrac{1}{2}\varphi. \tag{F.5}$$

Hence,

$$r_p = \frac{\sin\frac{1}{2}\beta}{\sin\frac{1}{2}\varphi}. \tag{F.6}$$

When $\varphi \approx 133.6°$, this formula sets the failure arc's radius as a function of f (or β). However, once the failure arc meets the drum surface, we must set $r_p = 1$ and $\varphi = \beta$, implying that

$$\mathrm{Od}_c = \frac{\sin^2(\beta/2)}{3\sqrt{2}\beta}. \tag{F.7}$$

The fill fraction at which the failure arc first reaches the drum surface is therefore when $\beta \approx 133.6°$, or $f \approx 0.2557$. The predictions above are shown in Fig. 11.

The results above can also be recovered in a different way: adopting the same failure geometry, we may compute the net vertical and horizontal force and torque on the rotating plug. This follows because the circular arc is an α−line. Therefore $p + 2\mathrm{Od}\ \theta$ is constant along the arc. The net forces and torque can then be computed in terms of three variables: the angle φ, Od_c and the pressure P_* at the lower end of the arc. In equilibrium, the net forces and torque must all vanish. The vanishing of the net torque implies a relation that can be reduced to (F.3), whereas the vertical and horizontal force balances give $\varphi = \tan(\varphi/2)$ and $P_* \approx 1.80\mathrm{Od}_c$.

The agreement between this calculation and the lower bound constructed from the power balance goes part way to proving that the circular failure arc must be the true solution in the plastic limit (which, according to the limit analysis of plasticity theory, follows if we verify that the lower bound using the velocity field meets the upper bound calculated from the stress field; [6,7]). The result is not completely established, however, because one must further construct an admissible stress field for all the plugged regions, which we have not done. Indeed, in the slender, low-fill-fraction limit, the rotating plug breaks over its thinnest sections, invalidating the result in (F.7) (*cf.* Fig. 12 and Appendix D).

Data availability

Data will be made available on request.

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